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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS3312-1 – FULL STACK DEVELOPMENT

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is a new element introduced in HTML5?
A) `<font>` B) `<center>`
C) `<article>` D) `<b>`
2. Which tag is used to define a navigation section in HTML5?
A) `<nav>` B) `<navigation>`
C) `<menu>` D) `<link>`
3. Which of the following is NOT a valid CSS selector?
A) `#id` B) `.class`
C) `*` D) `&element`
4. What is Bootstrap primarily used for?
A) Creating databases B) Front-end web development
C) Server-side scripting D) Network management
5. Which of the following is a JavaScript data type?
A) Integer B) String
C) Character D) Float
6. What does the `alert()` function do in JavaScript?
A) Displays a message to the user B) Logs a message to the console
C) Changes the background color D) Hides an element
7. What is the purpose of the `document.getElementById()` method in JavaScript?
A) To create a new element B) To access an element by its ID
C) To change the content of an element D) To delete an element
8. Which of the following is the correct way to include a JavaScript file in an HTML document?
A) `<script src="file.js">` B) `<script href="file.js">`
C) `<js src="file.js">` D) `<javascript src="file.js">`
9. Which of the following is NOT a valid use of PHP?
A) Server-side scripting B) Command-line scripting
C) Desktop application development D) Web browser development
10. What is the correct way to start a PHP script?
A) `<php>` B) `<?php>`
C) `<?script>` D) `<@php>`
11. What is the output of `echo 5 + "5";` in PHP?
A) 10 B) 55
C) 5+5 D) Error

12. Which PHP function is used to check the data type of a variable?
 A) getType() B) checkType()
 C) varType() D) dataType()
13. How do you create a database in MySQL using Node.js?
 A) db.create() B) CREATE DATABASE query
 C) db.createDatabase() D) mysql.createDatabase()
14. What does the SQL SELECT statement do?
 A) Deletes data from a table B) Selects data from a table
 C) Updates data in a table D) Creates a new table
15. How do you order the result of a SQL query in ascending order?
 A) ORDER BY column ASC B) SORT BY column ASC
 C) ORDER column ASC D) SORT column ASC
16. What does the LIMIT clause do in an SQL query?
 A) Limits the number of rows returned B) Limits the number of rows deleted
 C) Limits the number of rows updated D) Limits the number of columns displayed
17. How do you create a new database in MongoDB using Node.js?
 A) use database_name B) CREATE DATABASE database_name
 C) mongo.createDatabase(database_name) D) db.create(database_name)
18. Which method is used to insert a single document into a MongoDB collection?
 A) db.collection.insertOne() B) db.collection.add()
 C) db.collection.saveOne() D) db.collection.insertAll()
19. Which method is used to drop an entire collection from a MongoDB database?
 A) db.collection.dropCollection() B) db.collection.deleteAll()
 C) db.collection.drop() D) db.collection.removeAll()
20. Which Node.js module is used to connect with MongoDB?
 A) mysql B) mongoose
 C) mongo-client D) mongodb

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Develop a College Management System to create a Student Attendance Dashboard using Bootstrap. The dashboard should display a responsive table with student attendance records, following these requirements:
- Use Bootstrap table variants to indicate attendance status:
 - table-success for students with more than 75% attendance (Regular).
 - table-warning for students with 50% - 75% attendance (Needs Improvement).
 - table-danger for students with less than 50% attendance (Critical Warning).
 - The table should be striped, bordered, and hoverable for better readability.
 - Ensure the table is responsive, so it adapts to different screen sizes.
 - Add a "Last Updated" column to show the date of the last attendance update for each student.
- 8 L*3 1 2,3
- b) Develop a registration form using HTML5 that includes fields for name, email, password, and phone number. Ensure that:
- The email follows a proper email format.
 - The password requires at least 8 characters.
 - The phone number only accepts numeric values and must be exactly 10 digits long.
- Implement HTML5 form validation without using JavaScript.
- 8 L3 2 1,2

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|-------|---|---|----|---|-----|
| 2. a) | Explain various methods to add CSS to web pages along with its syntax. | 8 | L2 | 1 | 2,3 |
| b) | Design a button that: <ul style="list-style-type: none"> Changes color smoothly when hovered. Includes a subtle bouncing animation when clicked. Uses CSS transitions and keyframes for animation effects. | 8 | L3 | 1 | 2,3 |
| 3. a) | Explain the display possibilities in JavaScript with a code snippet for each. | 8 | L2 | 2 | 1,2 |
| b) | Build a countdown timer in JavaScript that: <ul style="list-style-type: none"> Takes user input for the number of seconds. Displays the remaining time dynamically, updating every second. Shows "Time's up!" when the countdown reaches zero. | 8 | L3 | 2 | 1,2 |

Unit – II

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|-------|--|---|----|---|-----|
| 4. a) | Explain types of arrays in PHP. | 8 | L2 | 3 | 1,2 |
| b) | Develop a Student Name Sorting System, where the system should take a list of student names and display them in ascending order (alphabetically). The names will be stored in an array, and you need to sort the names and display them.
Requirements: <ul style="list-style-type: none"> i. You will be given an array of student names (e.g., ["John Doe", "Alice Smith", "Bob Brown"]). ii. Sort the names alphabetically in ascending order. iii. Display the sorted list of names. | 8 | L3 | 3 | 1,2 |
| 5. a) | Explain output statements in PHP with a code snippet for each. | 8 | L2 | 3 | 1,2 |
| b) | Develop a program to demonstrate the scope of variables in PHP. | 8 | L3 | 3 | 1,2 |
| 6. a) | Explain any two looping statements in PHP along with an example for each. | 8 | L2 | 3 | 1,2 |
| b) | Create a simple webpage that uses the PHP GET method to retrieve and display user information from the URL parameters.
Create a URL that accepts the following parameters: <ul style="list-style-type: none"> name: A string representing the user's name. age: An integer representing the user's age. Use PHP to process the URL parameters and display the user's name and age on the webpage. If the parameters are not provided in the URL, display a message prompting the user to include their name and age. | 8 | L3 | 3 | 1,2 |

Unit – III

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|-------|---|---|----|---|-----|
| 7. a) | Compare the use of Node.js with other backend technologies like PHP in terms of handling high traffic applications. | 8 | L2 | 5 | 1,2 |
| b) | Develop a Node.js script that sets up a basic web server using the http module. | 8 | L3 | 3 | 1,2 |
| 8. a) | Explain various node.js events. | 8 | L2 | 5 | 1,2 |
| b) | Explain create database in Node JS MYSQL. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
